NATIONAL INSTITUTE OF TECHNOLOGY
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Mid Term Examination I: *Sept'2025* **MM:** *15*

Subject: *Design Analysis of Algorithms [CSPC201]* **Duration:** *50 mins*

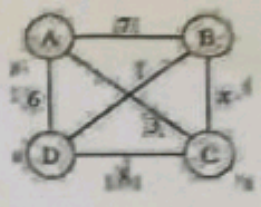
NOTE: 1. *The question paper contains* ***TEN*** *questions*
2. *All questions are compulsory.*
3. *Assume suitable data, if missing.*

Marks Distribution	
Q. 1, 2, 3, 9, & 10	1 Mark
other questions	2 Marks

Q1. Let G be any undirected graph with positive edge weights, and T be a minimum spanning tree of G. For any two vertices, u and v, let $d_1(u, v)$ and $d_2(u, v)$ be the shortest distances between u and v in G and T, respectively. Which ONE of the options is CORRECT for all possible G, T, u and v?

(A) $d_1(u, v) = d_2(u, v)$ **(B)** $d_1(u, v) \leq d_2(u, v)$ **(C)** $d_1(u, v) \geq d_2(u, v)$ **(D)** can't say

Q2. Find the maximum value of x such that the edge between the nodes **B** and **C** is included in every minimum spanning tree of the following graph (answer in integer).



Q3. Consider the following statements about an **almost complete binary tree** (with respect to its structure)

S1: There is only a single almost complete binary tree with n nodes.

S2: There are two distinct almost complete binary trees with n leaves.

Choose any one correct option from the following.

(A) Both are true **(B)** Both are false **(C)** S1 is true, S2 is false **(D)** S1 is false, S2 is true

Q4. Consider a directed graph **G = (V, E)**, where **V** = {0, 1, 2, …,100 } and $E = \{(i,j): 0 < j - i \leq 2,\ for\ all\ i,j \in V\}$. Suppose the adjacency list of each vertex is in decreasing order of vertex number, and depth-first search (DFS) is performed at vertex **0**. Find the number of vertices that will be discovered after vertex **50**.

Q5. Consider functions Function1 and Function2 expressed in pseudocode as follows:

Function1(n) {	Function2(n) {

while (n > 1)
for (i = 1 to n)
x = x + 1;
n =⌊ n / 2⌋;
}
	for (i = 1 to 100 * n)
	x = x + 1;
	}

Let $f_1(n)$ and $f_2(n)$ denote the number of times the statement "x = x + 1" is executed in Function1 and Function2, respectively. Which of the following statements is/are TRUE? (select one or more correct options)

(A) $f_1(n) \in O(f_2(n))$ **(B)** $f_1(n) \in o(f_2(n))$ **(C)** $f_1(n) \in \omega(f_2(n))$ **(D)** $f_1(n) \in O(f_2(n))$

Q6. Suppose a **binary search tree** with **1000** distinct elements is also an ***almost complete binary tree***. The tree is stored using the implicit array representation of binary heap trees. Assuming that the array indices start with **0**, find the index at which the **3rd** largest element of the tree is stored.

Q7. Assume that *disjoint-set data structure* is represented using a collection of doubly linked-lists with usual assumptions and the following operations: MAKE-SET, FIND-SET, and UNION.

Further, assume that the UNION procedure always appends the shorter list onto the longer and if the sets containing x_i and x_j have the same size, then the operation **UNION(x_i , x_j)** appends x_j's list onto x_i's list.

What will be the output returned by the FIND-SET operations in the following program.

```
for i= 1 to 16
    MAKE-SET(xi)
for i= 1 to 15 by 2 // i is incremented in a step of 2
    UNION(xi , xi+1)
for i= 1 to 13 by 4 // i is incremented in a step of 4
    UNION(xi , xi+2)
UNION(x1 , x5)
UNION(x11 , x13)
UNION(x1 , x10)
FIND-SET(x2)
FIND-SET(x9)
```

Q8. Breadth First Search (BFS) is started on a **binary tree** beginning from the root vertex. There is a vertex ***t*** at a distance four from the root. If ***t*** is the nth vertex in this BFS traversal, then find the maximum possible value of ***n***.

Q9. The numbers **1, 2, ..., n** are inserted in a **binary search tree** in some order. In the resulting tree, the right subtree of the root contains **p** nodes. The first number to be inserted in the tree must be:

(A) p **(B)** p+1 **(C)** n-p **(D)** n-p+1

Q10. With the usual implementation of the **MERGE-SORT(A, p, r)**, the operation of merge sort is performed on array **A = < 12, 3, 7, 9, 14, 6, 11, 2 >**. State the **truth value** (True or False) of the following statement:

" The elements 12 & 3 will be merged before dividing the elements 7 & 9 into their own subarray."